Thermodynamic Emergence Of Localized Observer Systems (TELOS)

Thermodynamic Trajectories, Observerhood, and Structural Ethics

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Abstract

Across physics, biology, artificial intelligence, and social systems, a striking set of regularities recurs. Cooperative structures often outlast dominating ones. Persistent deception becomes increasingly costly. Highly centralized control fails at scale. And value—where it exists at all—appears sharply localized rather than globally distributed.

The **TELOS** framework addresses these patterns by carefully separating two questions that are frequently conflated: *what is physically feasible* and *what is dynamically persistent*. Where the companion framework **ECHO** establishes thermodynamic and informational *feasibility ceilings* on sustained computation and organization, TELOS operates entirely within those ceilings, analyzing which *trajectories* can be maintained over long horizons and which inevitably collapse.

Central object. We analyze the dynamical landscape induced by the scale-invariant persistence functional

$$L^*(\pi, \Omega) = r_P(\pi) + c(\pi) + v(\pi) - \alpha_\Omega \, f_{\min}(\pi),$$

where long-run survival, internal predictive coherence, and causal empowerment contribute positively to persistence, while irreducible informational dissipation—scaled by an architecture-dependent overhead factor—imposes a thermodynamic cost. L^* *is not postulated as a universal objective.* Rather, it arises as a common gradient approximated by evolutionary, learning, and control processes as they operate under bounded resources and noisy environments.

Structural consequences. Within this framework, value is shown to be *strictly localized*: only self-modeling systems capable of maintaining predictive boundaries against noise can instantiate non-vanishing value gradients. Outside such bounded observers—whether physical, cognitive, or organizational—value density collapses to zero. Within viable boundaries, sustained deception is structurally unstable: maintaining incompatible internal and external models imposes irreducible inferential costs that increase dissipation and degrade long-run persistence. At larger scales, architectures whose coordination and control overhead grows superlinearly with system size are provably metastable, explaining the recurrent fragility of centralized coercive regimes despite their short-term efficacy.

Taken together, these results induce a structured *persistence landscape*. Certain regions sustain positive long-run viability, while others act as repellers in which coherence and empowerment collapse as dissipation diverges. Between these lie metastable traps—locally persistent regimes that fail under sufficient noise, scaling, or competition. *The framework therefore explains the transient success and eventual breakdown of destructive systems without invoking moral teleology or historical inevitability.*

From persistence to normativity. On this foundation, we define *functional caring* as the minimal tendency of a self-model to weight futures according to their capacity to preserve persistence and coherence. For self-modeling observers that encode their own continuation, functional caring is unavoidable. When multiple such observers coexist under shared thermodynamic constraints, patterns of interaction that erode the capacity of others to maintain self-models increase global dissipation, suppress predictive structure, and reduce long-run persistence for all involved. As a result, *preserving observerhood in others emerges as a necessary condition for sustained viability*, independent of moral preferences, cultural norms, or phenomenological assumptions.

This yields what we term a *thermodynamic ethics of persistence*: a **constraint-derived, observer-indexed ethical structure** grounded in dynamical gradients rather than moral axioms. It does not issue categorical imperatives, elevate values to properties of the physical substrate, or claim universal normativity beyond the domain of observers. Instead, it identifies where thermodynamic and informational constraints already do normative work by shaping which forms of agency, coordination, and valuation can continue to exist.

Scope and limits. TELOS introduces no new physical laws, does not solve human ethics, and makes no claims independent of observerhood. Its contribution is to clarify—*with physical precision and deliberate restraint*—what forms of value and normativity can arise once observers exist, and to constrain the space in which any viable ethics must operate.

1 Introduction and Scope

1.1 Motivation

Advanced artificial agents, large-scale social systems, and biological collectives all face the same fundamental constraint: persistence requires work. Maintaining internal structure, coordinating action, and preserving future optionality are not abstract virtues but physically costly processes governed by thermodynamics. As systems grow in complexity, the gap between what they can in principle compute and what they can sustainably maintain becomes increasingly consequential.

Recent discussions in artificial intelligence, collective governance, and ethics often oscillate between two unsatisfactory extremes. At one pole lie purely normative accounts that prescribe desirable outcomes without regard to physical feasibility. At the other lie purely descriptive accounts that enumerate constraints while denying any relevance to questions of agency, value, or survival. Neither position adequately addresses the practical problem faced by real systems embedded in finite worlds: given physical limits, which kinds of agents, policies, and organizations can persist?

This work is motivated by the need for a framework that links physical constraints to the long-run viability of value-bearing systems without smuggling inevitability, moral realism, or cosmological purpose into the analysis.

1.2 From Constraints to Trajectories

The companion work *ECHO* establishes thermodynamic ceilings on information processing and integrated structure. In particular, it relates the rate at which meaningful internal distinctions can be sustained to the energetic costs of erasure, coordination, and architectural inefficiency. These results impose hard bounds: some forms of growth, control, and intelligence are simply not physically admissible.

TELOS begins where ECHO ends. Rather than asking what is forbidden, we ask what remains possible. Given the constraints derived in ECHO, TELOS studies the *trajectories* of agents and collectives operating within those bounds, especially in open, multi-agent environments where resource acquisition, conflict, and coordination matter.

Where ECHO is a theory of limits, TELOS is a theory of regimes.

1.3 Unified Persistence Criterion

To evaluate long-run viability across disparate substrates and scales, TELOS introduces a persistence functional L^* that aggregates four structural quantities: (i) viability or survival rate, (ii) internal coherence, (iii) empowerment over future states, and (iv) dissipation or friction.

This functional is intentionally minimal. It does not encode any particular goal, preference, or moral axiom. Instead, it provides a common score by which different architectures and strategies can be compared in terms of their ability to sustain value-bearing internal structure over time, subject to physical cost.

Importantly, TELOS does not assume closed systems. Agents may acquire resources, extract energy, and externalize costs. However, such acquisition enters the framework only as a constraint and a dynamic driver; it does not appear as a primitive object of value. The central question remains persistence under load, not accumulation in the abstract.

1.4 Multi-Agent Interaction and Antagonistic Overhead

In multi-agent settings, strategies that increase short-term acquisition often introduce additional coordination, modeling, and control burdens. We refer to these burdens collectively as *antagonistic overhead*: the increase in effective dissipation associated with sustaining adversarial relationships.

Antagonistic overhead is not a moral penalty and does not preclude defection or competition. Rather, it captures a structural property of conflict-intensive strategies: as the number or intensity of antagonistic interactions grows, so too does the cost of maintaining internal coherence and control. This effect becomes especially salient in environments where multiple agents can coordinate in response.

Recognizing antagonistic overhead allows TELOS to distinguish between dominance that is merely transient and regimes that are dynamically viable in the long run.

1.5 Value Localization and Conditional Normativity

TELOS makes no attempt to derive value from physics. It assumes value where it is observed and treats it as a localized phenomenon supported by bounded, persistent systems. Agents need not care, and nihilistic or self-terminating strategies are not forbidden.

However, once value is instantiated—even minimally—its persistence is subject to constraint. TELOS therefore yields a form of *conditional normativity*: not prescriptions about what agents should value, but constraints on which strategies remain viable given that something is valued at all.

This position avoids both moral realism and fatalism. It neither guarantees desirable outcomes nor dismisses agency as epiphenomenal.

1.6 Admissible Futures and the Scope of the Analysis

A central result of this work is that the same physical framework permits qualitatively different long-run outcomes. Some regimes lead reliably to collapse or domination followed by failure; others admit stable, cooperative, and value-rich configurations. Neither outcome is inevitable, and no mechanism ensures convergence toward either.

Accordingly, TELOS should be read neither as a theory of cosmic justice nor as a counsel of despair. It is a map of a constrained landscape, identifying which futures are physically admissible, which are structurally fragile, and which depend on collective coordination to remain open.

The sections that follow formalize this framework, derive key consequences, and illustrate the resulting regime structure using analytical arguments and simplified simulations.

2 Epistemic Commitments and Relation to ECHO

2.1 Layer Separation: Feasibility vs. Trajectories

2.2 Status of Claims: Theorems, Conjectures, Interpretations

2.3 Assumptions and Minimal Ontology

3 Observers, Boundaries, and Value

3.1 Autonomous Systems and Predictive Boundaries

3.2 Observer Localization Theorem

3.3 Value as Asymptotic Persistence Gradient

3.4 The Anthropic Loop and Honest Tautologies

4 The Persistence Functional

4.1 Definition

We define the TELOS persistence functional as

$$L^*(\pi, \Omega) = w_P\, r_P(\pi) + w_C\, c(\pi, \Omega) + w_V\, v(\pi, \Omega) - w_F\, f(\pi, \Omega), \tag{1}$$

where π denotes a policy (or strategy), Ω an architectural instantiation, and $w_P, w_C, w_V, w_F > 0$ are fixed scaling coefficients used only to place heterogeneous terms on a common scale.

The terms are interpreted as follows:

- $r_P(\pi)$: *viability*, the expected persistence rate of the agent under policy π (e.g. probability of continuation per unit time),
- $c(\pi, \Omega)$: *coherence*, the degree to which internal representations, commitments, and control states remain mutually consistent,
- $v(\pi, \Omega)$: *empowerment*, the agent's effective control over reachable future states (optionality),
- $f(\pi, \Omega)$: *dissipation*, the total energetic and organizational cost required to execute π on Ω.

L^* is not a utility or reward function. It does not encode goals, preferences, or moral commitments. Instead, it evaluates the *structural viability* of value-bearing systems under physical constraint.

4.2 Universality and Minimality

Equation (1) is intentionally minimal. Each term is necessary:

- Without r_P, ephemeral systems dominate despite failing to persist.
- Without c, internally incoherent systems accrue hidden failure modes.
- Without v, systems can persist only as static remnants, not agents.
- Without f, physically impossible strategies appear admissible.

No additional positive terms are introduced. In particular, acquisition, resource extraction, or accumulation are *not* treated as intrinsic goods. Their relevance is mediated entirely through constraints on dissipation and the attainable values of r_P and v (see Section 5).

This minimality ensures that L^* applies uniformly across biological, artificial, and social systems, without privileging any particular substrate or objective.

4.3 Architectural Dependence and Inefficiency

Dissipation is decomposed as

$$f(\pi, \Omega) \; = \; \alpha_\Omega \, f_{\min}(\pi), \tag{2}$$

where $f_{\min}(\pi)$ is the minimal dissipation required to implement policy π at the relevant scale, and $\alpha_\Omega \geq 1$ is an architectural inefficiency factor capturing coordination overhead, control complexity, and non-ideal implementation.

This factorization separates unavoidable physical cost from costs imposed by design, organization, and interaction. Later sections examine how α_Ω increases under conflict and multi-agent antagonism.

4.4 Open Systems and Budget Constraints

Agents evaluated by L^* are not assumed to be closed systems. Let $A(\pi, \Omega, t)$ denote the rate of resource acquisition (e.g. energy, compute, organizational capacity) available over time. Viable execution requires the budget condition

$$f(\pi, \Omega, t) \;\leq\; \eta \, A(\pi, \Omega, t), \tag{3}$$

for some conversion efficiency $\eta \in (0, 1]$.

Importantly, A does not appear additively in L^*. Resource acquisition matters only insofar as it permits dissipation to be paid and empowerment to be exercised. This choice prevents trivial maximization of extraction from masquerading as long-run viability.

4.5 Interpretation

Maximizing L^* should be read as a selection criterion over policies and architectures:

> *Among strategies available to an agent that values its own persistence and optionality, those yielding higher L^* are structurally more viable under physical constraint.*

This interpretation is conditional rather than prescriptive. TELOS does not assert that agents must optimize L^*, only that sustained deviation from high-L^* regions results in failure to persist as a value-bearing system.

The remaining sections analyze how L^* behaves in open, multi-agent environments, where antagonistic overhead, coalition formation, and acquisition constraints reshape the persistence landscape.

Applicability Across Scales of Agency

It is important to emphasize that the analysis developed here does not presuppose civilization-scale organization or explicit deliberative agency. Natural ecologies already constitute richly dynamic, multi-agent systems operating under thermodynamic constraints, solving—via selection rather than intention—the same trade-offs between acquisition, dissipation, coordination, and persistence.

From this perspective, ecological systems represent the minimal realization of the TELOS framework: antagonistic overhead appears as predation and competition, cooperation appears as symbiosis and niche formation, and persistence emerges from sustained structural compatibility with environmental resource flows. Higher-order agents, institutions, and civilizations do not replace this dynamic; they embed within it, compress timescales, and internalize control.

Later sections refer to these baseline ecological dynamics under the label *Gaia-like regimes*, not as an idealized equilibrium, but as the lowest-complexity attractor in the persistence landscape.

5 Open Systems, Acquisition, and Resource Constraints

5.1 Agents as Open Thermodynamic Systems

Agents of interest in TELOS are not thermodynamically closed. Biological organisms, institutions, artificial intelligences, and coalitions all persist by continuously importing low-entropy resources and exporting waste. Their long-run behavior therefore cannot be analyzed without reference to resource acquisition and throughput.

However, openness does not relax physical constraints; it externalizes them. Acquisition enables continued dissipation, but does not eliminate its cost. Strategies that rely on increasing inflow merely shift the binding constraint from internal structure to external environment.

Accordingly, TELOS treats acquisition as a *necessary enabling condition* for persistence, not as a primitive object of optimization.

5.2 Resource Inflow and Viability

Let $A(\pi, \Omega, t)$ denote the resource inflow available to an agent executing policy π on architecture Ω at time t. This may include:

- metabolic energy (biological systems),
- electrical power and compute (artificial systems),
- economic, political, or organizational capacity (social systems).

Sustained viability requires that inflow compensates for dissipation:

$$\mathbb{E}_t[A(\pi, \Omega, t)] \;\geq\; \mathbb{E}_t[f(\pi, \Omega, t)]\,. \tag{4}$$

Violations of this condition imply rapid loss of viability, regardless of coherence or empowerment. Conversely, high inflow can temporarily mask architectural inefficiency or strategic fragility by operating near the dissipation ceiling.

This separation explains the empirical fact that highly inefficient or destabilizing systems can dominate for extended periods when supported by sufficiently large external resource bases.

5.3 Acquisition Does Not Alter the Persistence Criterion

Although acquisition enables greater dissipation, it does not alter the structure of the persistence functional. Increasing resource inflow cannot, by itself, increase L^* unless it improves r_P, c, or v

at commensurate cost.

This choice is deliberate. Treating acquisition as an additive good would trivialize persistence by making unbounded extraction a solution to all structural problems. In practice, strategies that maximize short-term acquisition often degrade coherence, introduce antagonistic overhead, or raise architectural inefficiency, thereby increasing f faster than they improve other terms.

Consequently, acquisition enters TELOS only through:

- feasibility constraints on dissipation,
- time-dependent effects on viability,
- interactions with architectural overhead in multi-agent contexts.

5.4 Temporal Asymmetry of High-A Strategies

High-acquisition strategies exhibit a characteristic temporal asymmetry. In the short term, increasing A expands the feasible region of execution, permitting aggressive growth, control, or dominance. Over longer horizons, however, such strategies tend to alter the agent's effective architecture by increasing coordination costs, conflict exposure, and control burdens.

In the absence of compensatory reductions in architectural inefficiency, this leads to a regime where

$$\frac{d}{dt}\,\alpha_{\Omega}^{\mathrm{eff}} \;>\; 0 \tag{5}$$

even if A remains large. Once the growth of effective inefficiency outpaces achievable acquisition, the agent is driven out of the viable region defined by the budget constraint.

This mechanism underlies the distinction between systems that are merely powerful and systems that are persistently viable.

5.5 Transition to Multi-Agent Dynamics

The consequences of open-system acquisition become most pronounced in environments containing multiple adaptive agents. Actions that increase A frequently do so by altering the environment or the payoffs available to others, eliciting strategic response.

In such settings, the cost of sustaining antagonistic relationships becomes a dominant contribution to architectural inefficiency. The next section formalizes this effect, introducing *antagonistic overhead* as a structural cost arising from conflict-intensive strategies in multi-agent systems.

6 Structural Theorems

6.1 Boundary Localization (Formal Proof)

6.2 Deception Instability

6.3 Overhead Cascade

6.4 Agency Ceiling

7 Repellers, Traps, and Instability

7.1 Repeller Sets

7.2 Metastable Traps

7.3 Finite Lifetime and Scale Effects

8 Persistence Regimes and Phase Structure

8.1 Regimes as Regions of the Persistence Landscape

Given the persistence functional defined in Section 4, together with open-system constraints and antagonistic overhead, the long-run behavior of agents can be understood in terms of *persistence regimes*. A regime corresponds to a region of parameter space characterized by typical relationships between acquisition, dissipation, coherence, empowerment, and antagonistic overhead.

Regimes are not agent types, moral categories, or equilibrium points. They are dynamical basins: classes of trajectories that exhibit similar long-term behavior under comparable constraints. Transitions between regimes are possible as parameters change, and no regime is guaranteed or forbidden *a priori*.

In what follows, we outline four qualitatively distinct regimes that emerge generically from the framework.

8.2 Regime I: Gaia-Like Systems

Gaia-like regimes correspond to low-acquisition, low-overhead systems whose persistence is primarily governed by compatibility with environmental resource flows. Natural ecologies exemplify this regime.

In such systems:

- antagonistic overhead is localized and diffuse (e.g. predation, competition),
- cooperation appears as niche formation and symbiosis rather than explicit coordination,
- coherence is maintained without centralized control,
- empowerment over distant futures is minimal.

These systems are thermodynamically robust and can persist over geological timescales. However, they possess limited capacity for deliberate coordination or defense against high-acquisition external agents. As such, Gaia-like regimes are stable but strategically vulnerable.

8.3 Regime II: High-Acquisition, High-Overhead Systems

In this regime, agents or collectives achieve dominance by aggressively increasing acquisition A, often at the expense of rising architectural inefficiency and antagonistic overhead.

Characteristic features include:

- sustained operation near dissipation ceilings,
- extensive control, enforcement, and monitoring costs,
- rapid scaling followed by brittleness,
- sensitivity to shocks or coordination failures.

These systems may dominate competitors for extended periods yet remain structurally fragile. Growth amplifies overhead faster than coherence or empowerment, eventually driving effective dissipation beyond sustainable acquisition.

This regime is often associated with collapse modes that are abrupt rather than gradual.

8.4 Regime III: Predatory Low-Overhead Systems

A third regime arises when an agent combines high acquisition with relatively low baseline inefficiency. Such systems can dominate isolated or weakly coordinated environments by maintaining low internal friction while externalizing costs.

However, this advantage depends critically on low antagonistic degree. When faced with multiple adaptive opponents or coordinated resistance, effective inefficiency increases rapidly, eroding viability.

Accordingly, predatory low-overhead systems are powerful in sparse or fragmented environments but become brittle in densely populated or coalitional settings.

8.5 Regime IV: Federated Systems

Federated regimes are characterized by selective cooperation among multiple agents combined with external strategic alignment. Internal antagonistic overhead is minimized through trust, shared modeling, and coordination, while external engagement remains conditional.

Key properties include:

- suppression of internal architectural inefficiency,
- amortization of defensive and coordination costs,
- positive-sum internal dynamics alongside selective defection externally,
- resilience to multi-front antagonism.

Federated systems are not inevitably stable nor universally optimal. Their viability depends on timely formation, sufficient acquisition to meet external threats, and mechanisms to prevent internal fragmentation. When these conditions are met, federations occupy regions of the persistence landscape inaccessible to isolated or universally antagonistic agents.

8.6 Phase Transitions and Path Dependence

Transitions between regimes are governed by changes in acquisition, inefficiency, antagonistic degree, and coordination capability. Because these variables interact nonlinearly, regime shifts are often path-dependent.

For example, delayed coordination can lock a system into high-overhead trajectories even when cooperative alternatives exist. Conversely, early federation may prevent later collapse by limiting growth in effective inefficiency.

The existence of multiple regimes implies that long-run outcomes are not determined solely by initial capability or intelligence. Small differences in interaction structure and timing can yield qualitatively different futures.

8.7 Summary

The regime structure described above formalizes the intuition that persistence depends not merely on power or efficiency, but on how acquisition, dissipation, and interaction scale together. The same physical constraints permit domination, collapse, or cooperative persistence, depending on how agents collectively shape their environment.

In the next section, we illustrate these regimes using simplified multi-agent simulations, highlighting the parameter regions in which federated trajectories become viable.

9 Antagonistic Overhead in Multi-Agent Systems

9.1 Conflict as a Structural Cost

In environments containing multiple adaptive agents, policies do not operate in isolation. Actions taken by one agent alter the strategic landscape faced by others, eliciting response. As a result, strategies that rely on sustained conflict or unilateral defection introduce additional costs beyond baseline execution.

We refer to these costs collectively as *antagonistic overhead*: the increase in effective dissipation associated with maintaining adversarial relationships. This overhead arises from several sources, including:

- increased modeling and prediction complexity,
- additional control and enforcement mechanisms,
- redundancy and defensive infrastructure,
- continual response to strategic counter-adaptation.

Antagonistic overhead is not a moral penalty, nor does it imply that conflict is irrational or forbidden. It is a structural consequence of operating in a competitive, adaptive environment.

9.2 Effective Architectural Inefficiency

We model antagonistic overhead as an increase in effective architectural inefficiency. Let k denote the agent's *antagonistic degree*: the number (or weighted intensity) of concurrently antagonistic relationships in which the agent is engaged. The effective inefficiency is then given by

$$\alpha_\Omega^{\text{eff}} = \alpha_\Omega \left(1 + \beta k\right), \tag{6}$$

where $\beta > 0$ parameterizes the marginal cost of additional antagonism.

Equation (6) is intentionally conservative. It assumes linear growth of overhead in k; superlinear scaling would strengthen subsequent results. The key requirement is merely that additional antagonistic fronts are not free.

9.3 Defection Without Universal Defectors

The introduction of antagonistic overhead does not rely on the existence of a universal defector or adversarial archetype. Agents may defect selectively, cooperate conditionally, or employ mixed strategies. The framework imposes no requirement that any agent behave altruistically.

Instead, antagonistic overhead accumulates as a consequence of strategic exposure. Policies that generate widespread hostility incur rising inefficiency even when they succeed in increasing short-term acquisition.

Formally, consider a family of policies $\{\pi_\theta\}$ parameterized by aggressiveness θ, such that short-term acquisition satisfies

$$\frac{dA}{d\theta} > 0 \tag{7}$$

over some range. If these policies also increase antagonistic degree, then beyond a threshold θ^*,

$$\frac{d}{d\theta}\, f(\pi_\theta, \Omega) \;>\; \frac{d}{d\theta}\big(r_P + c + v\big), \tag{8}$$

driving L^* negative despite continued access to high resource inflows.

This formalizes the intuition that defection can be locally profitable while remaining globally destabilizing.

9.4 Multi-Front Conflict and Brittleness

A central consequence of antagonistic overhead is the brittleness of strategies that rely on universal or near-universal antagonism. As k increases, marginal increases in acquisition must compensate not only for baseline dissipation, but also for growing inefficiency in coordination and control.

In open systems with finite environments or competing agents, this produces a characteristic failure mode: dominance through sheer throughput followed by abrupt loss of viability when effective dissipation exceeds achievable acquisition.

Importantly, this does not imply gradual decay. Systems operating near the dissipation ceiling are prone to rapid collapse once overhead grows beyond the capacity of available inflow.

9.5 Asymmetry Between Individuals and Coalitions

Antagonistic overhead introduces a structural asymmetry between isolated agents and coordinated coalitions. For an individual agent engaged in k adversarial relationships, overhead scales with k. For a coalition of n cooperating agents facing a common external adversary, internal antagonistic degree remains near zero, while defensive costs can be amortized across members.

This asymmetry does not guarantee coalition success. High-acquisition adversaries may still dominate. However, it creates regions of parameter space in which coordinated collectives achieve higher L^* than more powerful but universally antagonistic agents.

The implications of this asymmetry form the basis of the federation regimes analyzed in subsequent sections.

9.6 Summary

Antagonistic overhead provides a mechanism by which conflict-intensive strategies translate into increased dissipation without invoking moral assumptions or universal behavioral constraints. It preserves the permissibility of defection while explaining why strategies that generate widespread opposition tend to be dynamically brittle in multi-agent environments.

In the next section, we use this mechanism to classify persistent regimes and identify conditions under which cooperative federations constitute stable long-run attractors.

10 Conjectures and Asymptotic Structure

10.1 Strategy Convergence

10.2 Attractor Gradients

10.3 Pattern Portability

10.4 Resonance Amplification

11 Optional Interpretive Layer

11.1 Affective Correlates of Persistence Gradients

11.2 Why Phenomenology Is Non-Essential

12 Limitations and Open Problems

12.1 Bootstrap and Origins

12.2 Measurement of α_Ω

12.3 Non-Ergodicity and Ecological Closure

12.4 Aggregation Across Observers

13 Conclusion

13.1 What Has Been Established

13.2 What Remains to Be Tested

13.3 TELOS as a Research Program

Author Note. TELOS is offered not as doctrine, but as a structure for critique, simulation, and extension. Its success depends on where it fails.